

Project Brief: Working Document

Digital Audio Player • Rev 8

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1. Summary

A portable digital audio player designed from scratch by Alexander Loker: schematic, PCB, simulation, firmware and enclosure. The device is intended as a genuine end product (small, pleasant to use, and measurably good), and the project closes two deliberate skill gaps: custom MCU board design, and analog and mixed-signal design backed by simulation.

#	Objective	How it is satisfied
1	Build an audio player worth using	Portable microSD player, FLAC and MP3, single-ended 3.5 mm output tuned for IEMs, 2.4 in display.
2	Close the MCU-board knowledge gap	Custom ESP32-S3 board from the module up: power tree, boot straps, SDMMC, USB, I2S, I2C charger control.
3	Work with data converters	ES9038Q2M current-output DAC with a discrete I/V and filtering chain designed from scratch.
4	Work with analog signals	I/V conversion, differential summing, reconstruction filtering, gain structure, output impedance control, low-noise power conditioning.
5	Simulate where simulation earns its place	LTspice for the buck-boost, inverter and LDOs; signal-chain simulation where it informs a decision.

#	Objective	How it is satisfied
6	Learn	Each design decision documented and verified by measurement on the finished board.

Changes from Rev 6: summing stage values and simulation; output stage compensation, jack, ESD and plug detect; DC-fault protection by firmware monitoring of LDO power-good; maximum sample rate fixed at 176.4/192 kHz; power figures updated from Power Budget Rev 6 (runtime 7.8 h on derated capacity); resolved open questions removed.

2. Locked Scope

Decision	Choice	Rationale
MCU	ESP32-S3	USB-OTG device peripheral, 4-bit SDMMC, dual-core 240 MHz, octal PSRAM. ESP-IDF / FreeRTOS.
DAC	ES9038Q2M	Differential current output, which forces the full analog chain. Runs as I2S master with the DPLL off; all audio clocks derive from one local oscillator.
Clocking	Two tri-state oscillators on XI	22.5792 MHz (ECS-2520MVLC-225.792-CN-TR) and 24.576 MHz (Abracon ASTX-H11-24.576MHZ-T). Maximum sample rate 176.4/192 kHz.
Source	microSD, local files only	No Bluetooth, no streaming, no USB Audio Class.
Formats	FLAC 16/44.1 minimum, MP3	Decoded on the MCU to PCM. The DAC never sees a file format.
USB-C role	Charging and file transfer only	TinyUSB MSC exposes the microSD as a drive.
Output	3.5 mm TRS, single-ended, DC coupled	Same Sky SJ1-3535NS jack. DC coupling preserves the sub-1 Ω target; firmware mutes on a rail fault.
Load target	Simgot EW300 (28 Ω $\pm$ 15%, 119 to 121 dB/V); 16 to 32 Ω IEMs	Unforgiving of noise floor and output impedance.
Maximum output	0.6 V RMS (0.85 V peak)	114.6 dB SPL peaks on the EW300 (worst nozzle).
I/V and summing op-amp	TI OPA1612AIDRGR, 8-SON	Three dual packages: four I/V channels and two summing channels. 1.1 nV/ $\sqrt{\text{Hz}}$, 40 MHz GBW.
Output amp	TI OPA1622IDRCR, 10-pin VSON	Dual channel, unity-gain follower. Thermal pad connects to V-, not ground. EN pin used for shutdown.
Analog rails	$\pm$ 3.3 V	Headroom for 0.85 V peak output plus compensation drop.
Display	2.4 in IPS, 240 $\times$ 320, SPI	Glass ($\approx$ 43 $\times$ 60 mm) fits inside the battery footprint. ST7789 or ILI9341. No touchscreen.
Battery	Jauch LP605060JU+PCM, 1S1P 3.7 V, 1850 mAh	63.0 $\times$ 51.0 $\times$ 6.6 mm envelope, integrated PCM, 70 mm flying leads, no internal NTC.
Charger	TI BQ24250RGET	3 MHz synchronous buck with PowerPath, VQFN-24 4 $\times$ 4 mm. I2C plus standalone fallback. TS pin gives hardware JEITA. Charge at 500 mA (0.27C).
Analog pre-regulator	ADI LTC3440, SYS $\rightarrow$ 3.9 V	1.2 MHz synchronous buck-boost covering the full 3.0 to 4.2 V cell range. Feeds the LT3042 and the LT3462A inverter.
Negative pre-regulator	ADI LT3462AES6, +3V9 $\rightarrow$ -3.9 V	2.7 MHz dual-inductor inverter, integrated Schottky, TSOT-23-6.
Analog LDOs	ADI LT3042 / LT3093	$\pm$ 3.32 V, ultralow noise, current limit 125 mA on both.
Digital rail	ADI LTC3440, SYS $\rightarrow$ 3.3 V	600 kHz buck-boost, separate branch from SYS.

Decision	Choice	Rationale
Fuel gauge	Voltage-based, no coulomb counter	Load-compensated lookup table characterised from the spare cell.
EDA / CAD	Altium (AL-PN library), SolidWorks	Existing toolchain carries over.

3. Out of Scope

Item	Status	Note
4.4 mm balanced output	Dropped	Would require four amplifier channels.
Bluetooth / A2DP	Dropped	ESP32-S3 is BLE-only.
USB Audio Class	Deferred	Full-speed USB caps UAC1 around 24-bit / 96 kHz.
Coulomb counting	Rejected	Voltage-based gauging adopted.
BC1.2 / USB-C PD negotiation	Rejected	Input current limit set by EN1/EN2 or I2C. 5.1 k Ω CC pull-downs only.
Touchscreen	Rejected	Encoder and buttons are better in-pocket.
Line-in / recording	Deferred	Nothing in the design forecloses it.
DSP / parametric EQ	Deferred	Added in firmware once the signal path is trusted.
Bare-metal firmware	Rejected	Not where the learning value is.
Vented enclosure / active cooling	Rejected	Heat reaches the cell by conduction; venting would admit dust and moisture, and a fan is an acoustic noise source.

4. System Architecture

Signal chain: microSD $\rightarrow$ SDMMC (4-bit) $\rightarrow$ ESP32-S3 (decode to PCM in a PSRAM ring buffer) $\rightarrow$ I2S $\rightarrow$ ES9038Q2M $\rightarrow$ I/V conversion $\rightarrow$ differential summing and MFB low-pass $\rightarrow$ OPA1622 follower $\rightarrow$ 3.5 mm TRS. A local oscillator drives the DAC's XI pin, and the DAC generates BCLK and LRCLK for the ESP32-S3.

Power chain: Li-Po cell $\rightarrow$ BQ24250 (charge and PowerPath, SYS output) $\rightarrow$ LTC3440 to 3.9 V $\rightarrow$ LT3042 to +3.3 V, and LT3462A to -3.9 V $\rightarrow$ LT3093 to -3.3 V. The digital 3.3 V rail is a separate LTC3440 branch from SYS.

Grounding: a single GND net on one uninterrupted plane. Analog and digital are separated by placement and pour partitioning, never a split plane. +3V3_ANALOG and +3V3_DIG remain separate supply rails.

4.1 Digital Subsystem

- ESP32-S3 with octal PSRAM. Decode on one core, UI and I2S service on the other.
- SDMMC in 4-bit mode. Ring buffer sized against worst-case card latency. A load switch on card VDD allows power-gating; host pins are tri-stated before gating.
- UI: 2.4 in SPI TFT, EC11 encoder, tactile switches. Partial display updates only. Display off during playback with wake on interaction.
- Charger control over I2C, with GPIOs for EN1/EN2 and CE; STAT and INT into GPIOs.

4.2 Clocking

- The ES9038Q2M runs as I2S master with its DPLL off (synchronous mode, $MCLK = 128 \times f_s$). Every audio clock derives from one local oscillator, so ESP32-S3 clock jitter does not reach the conversion.
- Two oscillators, one per sample-rate family, joined at XI through 22 Ω resistors. The unused one is disabled through its tri-state OE pin; 10 k Ω pull-downs hold both off until firmware enables one.
- The DAC's clock-gear divider covers each family: $\div 1$, $\div 2$ and $\div 4$ give 176.4/88.2/44.1 kHz and 192/96/48 kHz.

4.3 Analog Chain

- I/V: four OPA1612 channels, +IN at 0 V, $R_f = 499\ \Omega$, $C_f = 1.5\ \text{nF}$ COG (corner 213 kHz, 0.038 dB loss at 20 kHz). The DAC's DC current is common to both legs and cancels in the summing stage.
- Summing: differential MFB Butterworth, gain 0.44, $f_0 = 69.4\ \text{kHz}$ (simulated $-3\ \text{dB}$ at 69.30 kHz). 499 Ω input, 220 Ω feedback, 681 Ω middle resistor, 13 nF to ground at each input node, 2.7 nF feedback.
- MFB was chosen over Sallen-Key: it attenuates and subtracts in one op-amp, keeps rolling off into the MHz region, and holds its inputs at a virtual ground.
- Chain loss at 20 kHz $\approx 0.068\ \text{dB}$; attenuation at 1 MHz $\approx 60\ \text{dB}$. COG/NPO capacitors throughout the signal path.

4.4 Output Stage

- OPA1622 unity-gain follower per channel with output-inclusive compensation: $R_{iso} = 4.7\ \Omega$, $R_{fb} = 499\ \Omega$ from the jack side, $C_f = 1.5\ \text{nF}$ from the op-amp side. $Z_{out} \approx 0.44\ \Omega$ at 20 kHz.
- Stable with IEM loads up to 1 nF of cable capacitance in simulation; 30% overshoot in the harsh 10 k Ω || 1 nF case.
- Enable divider (54.9 k Ω / 50 k Ω) gives 1.57 V from a 3.3 V GPIO, above the 0.95 V worst-case threshold, with a 2.6 ms ramp.
- DC-fault protection: firmware monitors the LT3042 and LT3093 power-good outputs and drives EN low on a rail fault.
- ESD122 on tip and ring at the jack. Plug detect on the jack's tip switch with a 100 k Ω pull-up, 10 k Ω series resistor and 10 k Ω tip bleed.

4.5 Power Tree

- LDO dropout sets the 3.9 V intermediate rail; the buck-boost covers the full cell range. The LT3042 holds high PSRR into the MHz region where switcher energy lives.
- Negative rail: LT3462A inverter, not a charge pump. Charge pumps were rejected on output resistance.
- The OPA1612 has no shutdown pin, so standby requires turning the analog branch off at the 3.9 V LTC3440.
- Switchers near the analog section: the charger at 3 MHz (only while plugged in), the LTC3440 at 1.2 MHz and the LT3462A at 2.7 MHz. Magnetic coupling and return currents are handled by placement, shielded inductors and return-path control.

4.6 Display Integration

- The design integrates a bare 2.4 in panel with an FPC connector. At this size the glass fits inside the battery footprint, so the display costs no additional volume.
- Requires a 0.5 or 0.3 mm pitch FPC connector and a constant-current boost backlight driver, placed away from the analog section with a shielded inductor.
- The panel must come from a vendor publishing its FPC pinout and backlight configuration.

5. Target Specifications

Parameter	Target	Comment
Output connector	3.5 mm TRS, single-ended	DC coupled, one jack
Rated load	Simgot EW300, 28 $\Omega \pm 15\%$	16 to 32 Ω IEMs supported
Output impedance	< 1 Ω , 20 Hz to 20 kHz	0.44 Ω calculated at 20 kHz
Maximum output	0.6 V RMS	$\approx 15\ \text{mW}$ into 23.8 Ω ; 114.6 dB SPL peak on the EW300
Output noise	< 3 μV RMS, A-weighted	Audible hiss floor is the practical limiter

Parameter	Target	Comment
THD+N	< -105 dB at 1 kHz, 32 Ω	The DAC is not the limiting element
Crosstalk	< -80 dB at 1 kHz	Single-ended shared ground return is the limit
Passband	Within 0.1 dB to 20 kHz	0.068 dB calculated for the I/V and summing chain
Formats	FLAC 16/44.1 and 16/48, MP3	24-bit and higher rates up to 192 kHz are a stretch goal
Battery life	≥ 7 h continuous playback	Power Budget Rev 6 predicts 7.8 h on derated capacity
Charge time	≈ 4.5 to 5 h at 500 mA	Including CV taper
Enclosure	$\approx 56 \times 75 \times 13.2$ mm	Driven by the $63 \times 51 \times 6.6$ mm cell envelope

6. Simulation Plan

Simulation is applied where it changes design decisions. The power tree's switchers and LDOs qualify; the charger's state machine does not, and its design work is datasheet arithmetic and thermal layout.

6.1 Power Tree (LTspice)

- LDO PSRR, output noise and output impedance: complete for the LT3093; LT3042 noise sweep complete.
- Buck-boost and inverter transient behaviour, ripple and inductor current: LT3462A complete.
- Start-up, UVLO and load-step response; backlight boost ripple and coupling path.

6.2 Signal Chain

- Summing-stage frequency response: complete (-3 dB at 69.30 kHz, 40 dB/decade).
- Output-stage stability into capacitive loads: complete.
- Remaining: output impedance versus frequency, I/V stage noise and stability against DAC output capacitance, full-path noise analysis, gain structure across the volume range.

6.3 Hand Calculation

- Charger inductor ripple and peak current, TS network values, charger junction temperature and charge time.

7. Risk Register

Risk	Impact	Mitigation
Battery 28-week lead time	Six-month stall if stock lapses	Purchase two units; the second is needed for discharge characterisation.
Runtime margin	6.5 h with the 20% design margin, below 7 h	Measure the display; reserves are MCU clock reduction and lower analog rails.
Charge time near the safety timer	Charging aborts early	Set the 9 h timer via I2C.
Charger switching noise	Noise floor degradation while plugged in	3 MHz sits inside the LT3042's PSRR range; placement, shielded inductor, minimal loop area.
LT3462A switch current at full scale	Negative rail clips	Simulated 0.282 A against a 300 mA minimum limit (6% margin).
LT3042 load at full scale	Current limit reached	≈ 106 mA peak against the 125 mA programmed limit.
OPA1622 VSON thermal pad at V-	Hand soldering not possible	Hot air or hotplate with a stencil; dedicated V- pour, never merged into ground.

Risk	Impact	Mitigation
DC-coupled output fault	Destroys an IEM	Firmware monitors LDO power-good and drives EN low. The LT3042 PG is high impedance in shutdown and does not flag a full loss of the +3V9 feed.
SD read latency	Audible dropouts	Ring buffer sized against worst-case latency; minimum occupancy logged.
Voltage-based gauging	Misleading battery indicator	Load compensation, measured lookup table, ADC calibration.
USB MSC and playback contending	Filesystem corruption	Enumeration stops playback and releases the volume before host mount.

8. Open Questions

#	Question
1	Volume control method: digital attenuation in the DAC, analog potentiometer, or digital potentiometer ahead of the output stage. Affects the noise floor at low listening levels.
2	Display panel vendor, with a published FPC pinout and backlight configuration.
3	Measurement equipment: a -105 dB target needs an analyser floor below it.

9. Revision Log

Rev	Date	Change
6	2026-09-24	Clocking, I/V stage and grounding decisions; charging section, revision roadmap and simulation-limits section removed.
7	2026-09-27	Reformatted. Summing, output stage, jack, ESD and DC-fault decisions; maximum sample rate; power figures from Power Budget Rev 6; resolved open questions removed.
8	2026-09-27	Scope-creep risk removed.

10. Sources

1. [ES9038Q2M Datasheet v1.4 \(ESS Technology\)](#)
2. [ECS-2520MVLC Datasheet \(ECS Inc.\)](#)
3. [ASTX-H11 Datasheet \(Abracon\)](#)
4. [OPA1612 Product Page and Datasheet \(Texas Instruments\)](#)
5. [OPA1622 Datasheet, SBOS727B \(Texas Instruments\)](#)
6. [Analog Engineer's Circuit Cookbook: Amplifiers, MFB Low-Pass Filter \(Texas Instruments\)](#)
7. [SJ1-353XNS Series Datasheet \(Same Sky\)](#)
8. [ESD122 Product Page and Datasheet \(Texas Instruments\)](#)
9. [bq2425x Product Page and Datasheet \(Texas Instruments\)](#)
10. [TPD2E2U06-Q1 Product Page and Datasheet \(Texas Instruments\)](#)
11. [LTC3440 Product Page and Datasheet \(Analog Devices\)](#)
12. [LT3462/LT3462A Datasheet \(Analog Devices\)](#)
13. [LT3042 Datasheet \(Analog Devices\)](#)
14. [LT3093 Datasheet \(Analog Devices\)](#)